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Recall

1. V.S. = abelian group + S.M / $\mathbb{F}$ (scalar multiplication)
2. |Basis| = dimension $\begin{cases} \text{finite} \\ \text{infinite} \end{cases}$
3. $V \times W$ — Product $\dim(V \times W) = \dim V + \dim W$
4. Linear maps $V \xrightarrow{\sigma} W$ s.t. $\sigma(xv + x'v')$

   $v \xrightarrow{\sigma} \sigma(v)$ $= x\sigma(v) + x'\sigma(v')$

   $Hom_{\mathbb{F}}(V, W) \subseteq W^V$
5. Isomorphism

   surjection = surjective transformation ($Im\,\sigma = W$)

   Injection ($Ker\,\sigma = 0$)

   isomorphism = sur + inj = bijection

RK. $\exists\,\sigma \in Hom(V, W)$ an isomorphism $\Leftrightarrow \dim W = \dim V$

Linear Expansion

suppose $v_1, \dots v_n$ a basis of V

$w_1, \dots w_m$ a basis of W

$v = (v_1 \cdots v_n)X$ $V \longrightarrow W$

$\sigma: v_i \longrightarrow \sigma(v_i)$

$v + v' \longrightarrow w + w'$

$(v_1, \cdots v_n)(X + X') \longrightarrow \sigma(v + v') = \sigma(v) + \sigma(v')$

$= \sigma((v_1, \cdots v_n)X) + \sigma((v_1, \cdots v_n)X')$

$= (\sigma(v_1), \sigma(v_2) \cdots \sigma(v_n))X + (\sigma(v_1) \cdots \sigma(v_n))X'$

$= (\sigma(v_1), \cdots \sigma(v_n))(X + X')$

Natural Question

If σ is an isom then σ^{-1} is also an isom.

Pf. $\sigma^{-1}(w + w') = \sigma^{-1}(w) + \sigma^{-1}(w')$

$\exists v, v' \in V$ s.t. $w = \sigma(v)$, $w' = \sigma(v') \Rightarrow \sigma^{-1}(w) = v$, $\sigma^{-1}(w') = v'$

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$\sigma^{-1}(w+w') = \sigma^{-1}(\sigma(v)+\sigma(v'))$

$= \sigma^{-1}(\sigma(v+v'))$

$= (\sigma^{-1}\sigma)(v+v')$

$= v+v'$, $\sigma^{-1}(w)+\sigma^{-1}(w')$

so $\forall\ \sigma \in Hom(V, W)$ TFAE:

1. σ is an isomorphism
2. σ^{-1} is an isomorphism
3. σ is bijection

Furthermore if $V = W$, $\dim V = n < \infty$ $Hom_F(V,W) = End_F V$

[End = endowmorphism 自同态]

· Let $\sigma \in End\ V$. Then σ is an isomorphism $\Leftrightarrow$ σ is injective $\Leftrightarrow$ σ is surjective

(3) The second dimension formula: Let $\dim V = n$ $\sigma \in Hom(V, W)$

Then $\dim \ker v + \dim Im\, \sigma = n$

Lemma. $Ker\,\sigma$ and $Im\,\sigma$ are naturally equipped vector space structure

$\sigma(v_1+v_2) = \sigma(v_1)+\sigma(v_2) = 0+0 = 0$

$\sigma(x\, v_i) = x\,\sigma(v_i) = 0$

(criterion)
Lemma A subset U of V is a subspace iff U is closed under "+" and "·"

pf. Claim $\sigma(v_{s+1}), \dots \sigma(v_n)$ form a basis of $Im\,\sigma$

$\forall\, w \in Im\,\sigma$, $\exists\, v \in V$ s.t. $w = \sigma(v) = \sigma(x_{s+1}v_{s+1} + \cdots + x_n v_n)$

$= \sigma(x_{s+1}v_{s+1}) + \cdots + \sigma(x_n v_n)$

$= x_{s+1}\sigma(v_{s+1}) + \cdots + x_n\sigma(v_n)$

$0 = \sigma(x_{s+1}v_{s+1}, \cdots, x_n v_n)$, $x_{s+1} = \cdots = x_n = 0$

EX. $\mathbb{R}[x] \xrightarrow{\sigma:\int} \mathbb{R}[x]$

$f(x) \longrightarrow \int_0^x f(t)dt$ σ is injective but not surjective

$End\,\mathbb{R} = \begin{cases} \sigma : \mathbb{R} \to \mathbb{R} \\ \sigma(x) = kx \end{cases}$

$\{\sigma_k : x \mapsto kx \mid x \in \mathbb{R}, k \in \mathbb{R}\}$

$+ : \sigma_k + \sigma_{k'} : x \longmapsto (k+k')x$

$\cdot : \lambda\sigma_k : x \longmapsto \lambda k x$